Ipek Naz Obek

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Education

B.A., Cognitive Science University of Pennsylvania, Philadelphia, PA May 2025
Concentration: Computation and Cognition
Minors: Data Science, Computational Neuroscience
Summa Cum Laude; Distinction in Cognitive Science
Benjamin Franklin Scholar; Certificate in French
Honors Thesis: *Inferring Level Progress in Goal-Directed and Open-Ended Tasks*
Advisors: Dr. Alison Gopnik, Dr. Sharon Thompson-Schill, Annya Dahmani
Relevant Coursework: Programming Languages, Probability, Data Mining, Statistical Inference, Calculus I & II, Linear Algebra, Cognitive & Computational Neuroscience, Judgment & Decisions, Behavioral Economics, Developmental Psychology

Research Experience

Computation, Learning, and Development Lab Boston University, Boston, MA Aug 2025–present
Research Technician and Lab Manager
PI: Dr. Kate Nussenbaum

- Lead the design and implementation of online behavioral studies investigating goal pursuit strategies and persistence in children, adolescents, and adults.
- Design and program browser-based behavioral experiments using jsPsych, PsychoPy, and Qualtrics.
- Analyze trial-level behavioral data using mixed-effects models in R to characterize developmental and individual differences in learning and choice.
- Implement computational models of sequential decision-making in Python to test competing accounts of human exploration and value-guided learning.
- As the lab's first lab manager, established experimental, recruitment, and data-collection infrastructure, recruiting over 1,000 participants into the lab's database for developmental studies.
- Organize a biweekly multi-lab journal club bringing together researchers to discuss current papers.

Gopnik Cognitive Development and Learning Lab UC Berkeley, Berkeley, CA Jun 2024–May 2025
Research Assistant
PI: Dr. Alison Gopnik; Advisor: Annya Dahmani

- Designed a novel behavioral paradigm using Procgen environments to investigate how people infer learning progress in goal-directed and open-ended tasks.
- Developed computational accounts of learning-progress inference and compared human judgments with reinforcement-learning agents and large language models.
- Designed and analyzed experiments examining how people construct and evaluate curricula for complex, multi-step tasks.
- Supported by the Pinkel Research Fund, Benjamin Franklin Research Grant, and Career Services Research Grant; received the Benjamin Franklin Talk Prize.
- Recruited families on-site and conducted in-person behavioral testing with children in a museum setting, administering experimental protocols and collecting developmental data.

Thompson-Schill Lab — University of Pennsylvania, Philadelphia, PA — Jan 2024–May 2025
Research Assistant
PIs: Dr. Sharon Thompson-Schill, Dr. Anna Jenkins

- Conducted honors thesis research, jointly advised across the Thompson-Schill and Gopnik Labs, on human and AI representations of learning progress in goal-directed and open-ended environments.
- Participated in weekly joint lab meetings, discussing current research and literature with faculty and lab members.

Center for Neuroengineering and Therapeutics — University of Pennsylvania, Philadelphia, PA — Jun–Sep 2023
Research Assistant
PI: Dr. Brian Litt; Advisor: Dr. James Gugger

- Investigated relationships between traumatic-brain-injury lesion characteristics and subsequent post-traumatic epilepsy using multimodal neuroimaging data.
- Developed Python-based quantitative and machine-learning analyses relating lesion volume and location to clinical outcomes.
- Built an automated pipeline for MRI lesion segmentation and volumetric analysis, reducing reliance on manual annotation workflows.

Honors & Awards

Distinction in Cognitive Science — University of Pennsylvania — 2025
Awarded for completing an honors thesis, "Inferring Level Progress in Goal-Directed and Open-Ended Tasks," and for academic excellence in the major. Thesis advised by Alison Gopnik and Annya Dahmani (UC Berkeley Psychology) and Sharon Thompson-Schill (Penn Psychology).

Summa Cum Laude — University of Pennsylvania — 2025
Graduated with highest Latin honors from the College of Arts & Sciences.

Benjamin Franklin Talk Prize — Benjamin Franklin Scholars, University of Pennsylvania — 2025
Recognized for the best senior research presentation among the 2025 cohort of Benjamin Franklin Scholars, Penn's interdisciplinary honors program.

Pinkel Research Fund — University of Pennsylvania — 2025
Research funding awarded to support my senior honors thesis on curriculum learning in humans and machines.

Lucid Communication Challenge, Engineering and Physical Sciences Talk — Penn National Research Conference — 2024
Selected from a field of applicants at the National Research Conference at Penn to present my honors thesis research as a talk.

Benjamin Franklin Research Grant — Benjamin Franklin Scholars, University of Pennsylvania — 2024
Summer research funding awarded by the Benjamin Franklin Scholars Program; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

Career Services Research Grant — Penn Career Services — 2024
Competitive summer funding awarded by Penn Career Services; supported my research in the Gopnik Cognitive Development and Learning Lab at UC Berkeley.

MindCORE Fellowship — MindCORE, University of Pennsylvania — 2023
Selected for a fully funded 10-week summer research program in cognitive science and neuroscience at Penn's hub for the integrative study of the mind.

Undergraduate Research Mentoring Award — Penn Undergraduate Research Mentoring Program — 2023

Fellowship awarded through Penn's Undergraduate Research Mentoring Program, which funds first- and second-year students to conduct summer research with a faculty mentor.

Dean's List — University of Pennsylvania — 2021–2025

Awarded in every academic year for a GPA of 3.7 or higher.

Benjamin Franklin Scholar — University of Pennsylvania — 2021–2025

Member of Penn's interdisciplinary honors program in the liberal arts and sciences.

Manuscripts in Preparation

Obek, I. N., Fu, J., Holton, E., & Nussenbaum, K. (2026). *Examining the development of goal commitment*. In preparation.

Presentations

- **Obek, I. N.**, Fu, J., Holton, E., & Nussenbaum, K. (2026, September). *Examining the development of goal commitment* [Poster]. Flux Society for Developmental Cognitive Neuroscience, La Jolla, CA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Poster]. Curiosity, Information Seeking, and Exploration Conference, Brown University, Providence, RI.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2025, October). *Automated curriculum learning in children and machines* [Talk]. Big Team Science Conference.
- **Obek, I. N.** (2025, April). *Machines that think like kids* [Talk]. Benjamin Franklin Scholars, University of Pennsylvania, Philadelphia, PA. Awarded the Benjamin Franklin Talk Prize.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Talk]. Lucid Communication Challenge, Engineering and Physical Sciences session, Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Dahmani, A., & Gopnik, A. (2024, September). *Automated curriculum learning in children and machines* [Poster]. Penn National Research Conference, Philadelphia, PA.
- **Obek, I. N.**, Gugger, J., Narang, S., Vasireddy, M., & Zhao, C. (2023, August). *Understanding the link between TBI-induced lesions and epilepsy* [Poster]. MindCORE Research Fellowship Conference, Philadelphia, PA.

Teaching

Neuroscience and Society — Teaching Assistant, Penn Undergraduate Program in Neuroscience — 2024–2025

Instructor: Dr. Sharon Thompson-Schill

Supported students through weekly office hours, in-class instruction, and group activities, and provided individualized academic guidance.

Introduction to Cognitive Science — Teaching Assistant, Penn Undergraduate Program in Cognitive Science — Fall 2023

Instructor: Dr. Russell Richie

Led weekly recitations and office hours for 20+ students, developed instructional materials, and guided students through course concepts and assignments.

MATH 114: Calculus II — Tutor, Penn Weingarten Tutoring Center — Fall 2022

Tutored first-year students for 4 hours per week, providing individualized support with mathematical problem-solving and study strategies.

Mentorship

Benjamin Franklin Scholars Mentor Program — Peer Mentor, University of Pennsylvania — 2024–2025

Mentored 7 Benjamin Franklin Scholars in their academic and social transition to Penn through regular individual outreach and community-building events.

Center for Undergraduate Research and Fellowships — Research Peer Advisor, University of Pennsylvania — 2024–2025

Advised undergraduates pursuing research opportunities and organized research programming, including an international faculty symposium on the experiences and challenges of international academics.

Pioneer Academics — Program Coordinator and Advisory Board Member — 2022–2025

Advised 100+ students through the research and writing process and coordinated research seminars on effective research practices and faculty mentorship.

Young Guru Academy — Alumni Mentor — 2019–present

Mentor Turkish high school and university students pursuing international education and STEM careers. Led STEM outreach across 5+ cities in rural Anatolia, training teachers and students and delivering hands-on science education to underserved communities; presented at YGA's annual summit to ~20,000 attendees.

Leadership & Service

Cognitive Science Student Association — Founder & President, University of Pennsylvania — 2024–2025

Founded and led Penn's Cognitive Science Student Association, organizing faculty talks, research seminars, mentoring, and networking initiatives connecting students with faculty, alumni, and industry professionals.

The DREAM Program — Recruitment Chair, Philadelphia, PA — 2023–2025

Led volunteer recruitment and weekly educational programming for preschool-aged children in West Philadelphia, including hands-on science activities designed to foster curiosity, collaboration, and confidence in learning.

Turkish Student Association — President, University of Pennsylvania — 2022–2025

Directed a 50-member organization and community initiatives, including raising $8K for earthquake-relief efforts.

Assembly of International Students — Social Committee Member, University of Pennsylvania — 2021–2025

Organized social and cultural programming to foster community among Penn's international student population.

The Daily Pennsylvanian — Photographer, University of Pennsylvania — 2021–2025

Photographed campus events and contributed visual content for multimedia stories at Penn's student newspaper.

Skills

Programming: Python, R, JavaScript, MATLAB, SQL, Java, OCaml

Research tools & platforms: jsPsych, PsychoPy, Qualtrics, Prolific, Pavlovia, Git/GitHub, Jupyter, LaTeX, AFNI

Certifications: CITI Training (Biomedical Responsible Conduct of Research; Human Subjects Research; Social-Behavioral Research with Children)

Languages: Turkish (native), English (fluent), French (fluent)

References

Dr. Kate Nussenbaum
Assistant Professor, Department of Psychological and Brain Sciences
Boston University
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Dr. Alison Gopnik
Professor of Psychology
University of California, Berkeley
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Dr. Sharon Thompson-Schill
Christopher H. Browne Distinguished Professor of Psychology; Co-Director, Center for Cognitive Neuroscience
University of Pennsylvania
sschill@psych.upenn.edu

Dr. Russell Richie
Director of Strategy and Impact
Progress and Poverty Institute
Formerly Associate Director, Cognitive Science and MindCORE, University of Pennsylvania
drussellmrichie@gmail.com

Annya Dahmani
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